

prior information to determine which network is the most likely to lie behind a given observation, via the Bayesian posterior distribution $P(\mathbf{A} \mid \mathbf{D}) = \frac{P(\mathbf{D}|\mathbf{A})P(\mathbf{A})}{P(\mathbf{D})}$, where $\mathbf{D}$ is the observing data, $\mathbf{A}$ is the underlying network $\mathbf{A}^{(17)}$. Especially, the problem of bipartite graph inference is a transitional issue. A bipartite graph is a graph in the form $\mathcal{G} = \{\mathcal{U}, \mathcal{V}, \mathcal{E}\}$ such that its vertices can be divided into two disjointed sets $\mathcal{U}$ and $\mathcal{V}$ and there is no edge between vertices of the same set. The overall connectivity is usually described by incidence matrix $\mathbf{B}_{|\mathcal{U}| \times |\mathcal{V}|} = \{b_{ij}\}$ and its transposed form $\mathbf{W} = \mathbf{B}\mathbf{B}^\top = \{w_{ij}\}$. To measure the possibility of the existence of one latent edge, researchers introduce the metric $\pi_{i,j}$ as attraction coefficient of i and j . Given the original π_{ij} , the particular Bayesian application is $P(\pi_{ij} \mid w_{ij}, x_{ij}) = \frac{P(w_{ij}|\pi_{ij}, x_{ij})P(\pi_{ij})}{\int_0^1 P(w_{ij}, \pi_{ij}) dx}$, where $x_{ij} = \sum_{k=1}^{|\mathcal{V}|} \mathbf{1}\{b_{ik} \vee b_{jk}\}$. And we use this Bayesian equation to update the prior distribution of π_{ij} by the posterior $\pi_{ij} \mid w_{ij}, x_{ij}$. But a common drawback for maximum likelihood and Bayesian inference methods is that the complexity of underlying network structure explosively raises the computational cost, which may not make maximum likelihood approaches possible for massive real-world network datasets

Inspired by the idea of maximum and Bayesian methodology, we begin to introduce similar statistical inference approach to other spreading fundamental models, e.g., branching process. Branching process is one of the most classical stochastic models. It is widely used in epidemic theory, providing a simple way for approximation to spread process. Geometrically, we can consider a tree structure generated by branching process and analyze its statistical attributes to build foundation for the following inference of missing links. In present work on branching process, common aim of statistical inference is to estimate the parameters like offspring distribution rather than the measure of latent branching links which is similar to attraction coefficient in bipartite network.

We structure this work as follows. In Section 2, we introduce preliminary knowledge about branching processes and give a description of our model, which can be decomposed into another description framework of branching process and how to implement Bayesian method to the reconstruction problem while ensure the compatibility. In Section 3, we derive algorithms from previous theoretical formulation. In Section 4, numerical simulations are carried out for better understanding of different algorithm performances and the impact of branching properties on reconstruction accuracy. In last section, we conclude with a short discussion and propose our expectations for future applications and theoretical extensions.

2 MODEL FORMULATION

SIR model, the most famous and straightforward epidemiological model of how infection spreads in a closed population, is based on dividing the population in either of three states: susceptible, infectious, and recovered.⁽¹⁹⁾ Ample evidence of epidemic outbreaks is investigated with its assist.

But, due to the lack of an expression for transition probabilities, the mechanistic analysis can be troublesome. It's hard to derive an explicit solution. In previous research⁽⁸⁾, it is proved that BP can make a fair approximation of SIR model by mimicking the interaction effect of infection over short time intervals. It gives us the reason to take the framework of branching process as the cornerstone for reconstruction.

Our objective is to reconstruct the spreading system when partial observation occurs. Some missing individuals need to be reconnected to their parents, which means reconnect unobserved links. In following subsections, we will try to formulate the branching process framework and how to justify whether a reconstruction inference is reliable or not.

2.1 PRELIMINARIES

In this section, we review the definition and normal representation form of branching process. The branching process is one of the most typical and practical models in stochastic area, can be used to describe a wide range of spreading phenomena. Let start with Galton-Watson Branching Process (GWBP) model. It will branch with the following rules: every individual reproduces only once at the end of its unit length lifetime. And we suppose that reproduction or infection number of each individual is a non-negative integer random variable ξ , where its distribution $\mathcal{D}$ is the so-called offspring distribution.

For simplicity, let random variable Z_i denote the number of individuals in the i -th generation, X_{ij} denote the corresponding ξ belonging to the j -th individual of i -th generation. Conventionally, set the original population Z_0 to be 1, then for a branching process, the relationship between $\{Z_i\}$ and $\{X_{ij}\}$ can be represented as:

$$Z_{i+1} = \sum_{j=1}^{Z_i} X_{ij}.$$

It is easy to verify that the expected value of Z_i can be written as $\mathbb{E}[Z_i] = \mathbb{E}[\xi]^i$. Now consider a sufficiently large enough i , after i generations, a branching process would either survive or extinct. For more precise calculation of the probability of extinction, we define the generating function $f(\rho)$ as:

$$f(\rho) = \sum_i P(\xi = i) \rho^i.$$

The solution ρ of the fixed point equation $f(\rho) = \rho$ over interval $[0, 1]$ is the probability of extinction. By this equation, we get conclusions about extinction probabilities: when $E(\xi) < 1$, the probability of ultimate extinction is 1; when $E(\xi) = 1$, it is a critical case; when $E(\xi) > 1$, both extinction and survival are possible. In our work, we always ensure $E(\xi) > 1$ to guarantee the survival of process. Furthermore, if age-dependence is considered, the Galton-Watson BP will become Bellman-Harris BP only by replacing the default unit life-time of individual with random length of time from a given distribution.

To model multiwave pandemic dynamics, Crump-Mode-Jagers BP is introduced, where a single individual is allowed to reproduce multiple times during life-time instead of only once when death occurs. The reproduction mechanism is considered as a point process⁽¹⁰⁾.

2.2 TRANSPOSITION ON OFFSPRING DISTRIBUTION

As previously discussed, a GWBP can be fully represented by $\{X_{ij}\}$, the collection of random variables that represent the reproduction numbers per individual. Suppose the basic parameters of a GWBP are known (depth, offspring distribution, original population), for i -th generation, given an inference $\mathbf{x}_i = (x_1, x_2, \dots, x_{n_i})$ for $\mathbf{X}_i = (X_1, X_2, \dots, X_{n_i})$, a metric $\Pr[\mathbf{X}_i = \mathbf{x}_i]$ for inference is used to evaluate this inference. The calculation is

$$P(\mathbf{X}_i = \mathbf{x}_i) = P(X_1 = x_1, \dots, X_{n_i} = x_{n_i}) = \prod_{j=1}^{n_i} P(X_j = x_j) = \prod_{j=1}^{n_i} p_{x_j},$$

where X_{ij} and x_{ij} are shortly denoted as X_j and x_j , n_i is the population of i -th generation, and $\{p_k; k = 0, 1, \dots\}$ is the distribution function of random variable ξ .

But in real world, restricted by the imperfectness of observation, it may be hard to figure out detail